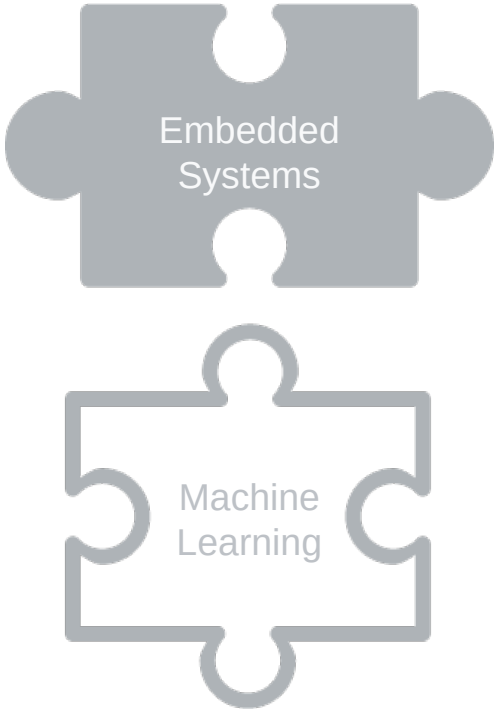
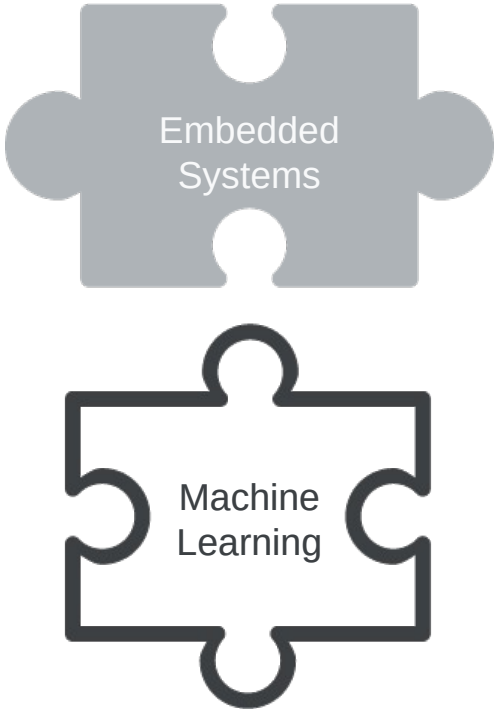
What are the Challenges for TinyML?

Part C

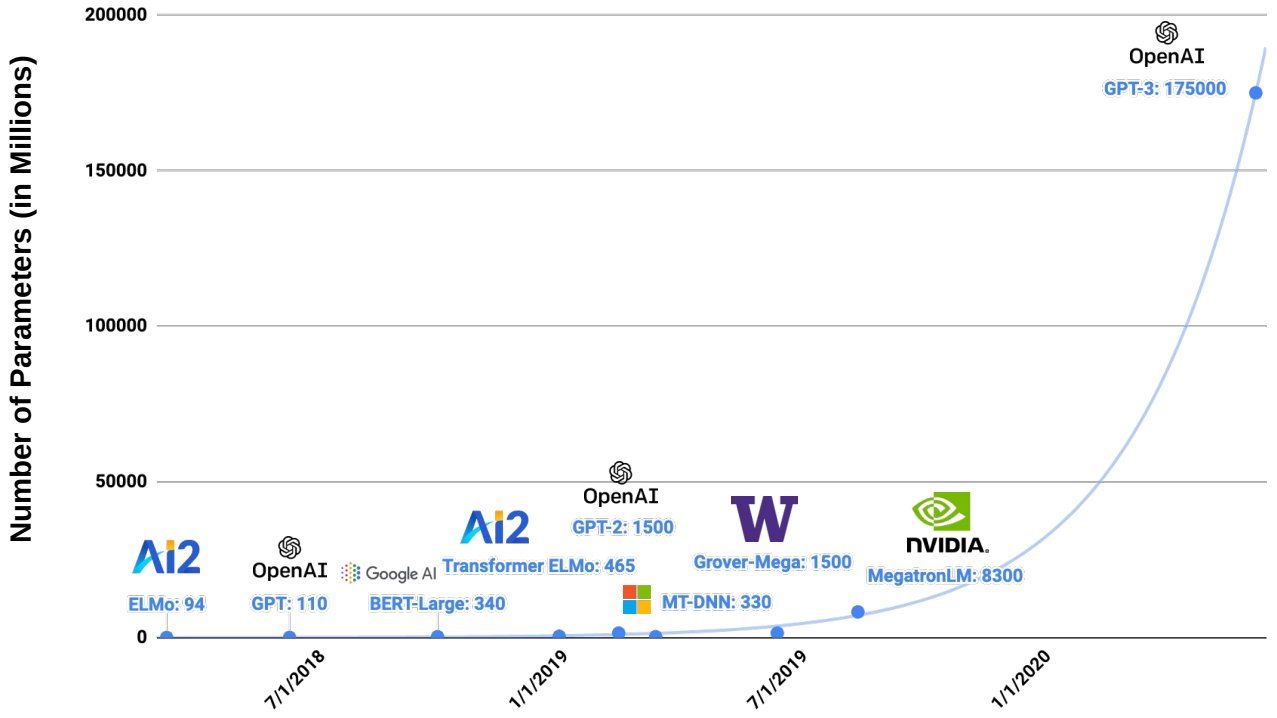


TinyML

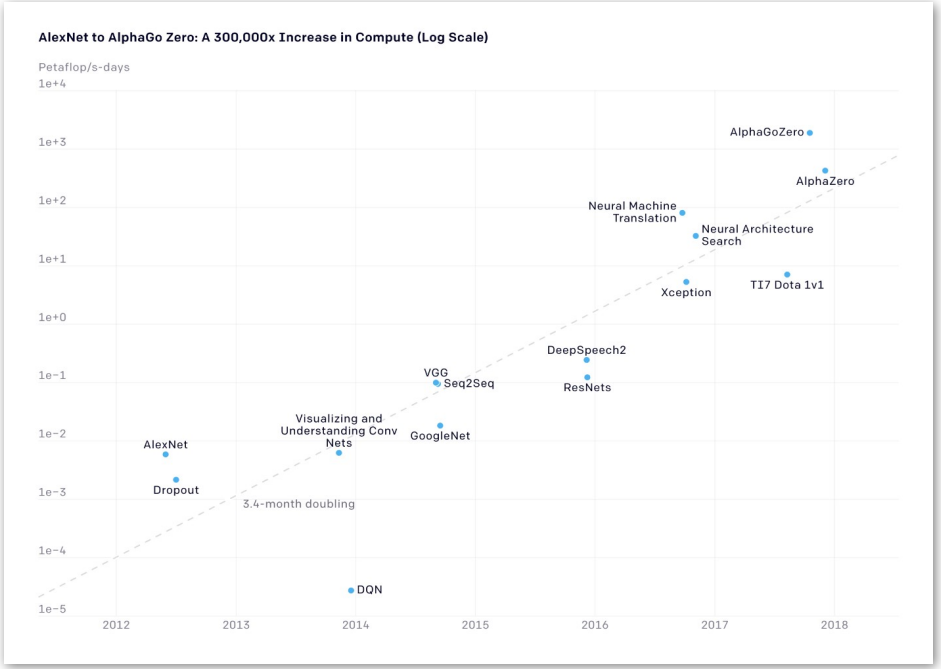


TinyML

ML Model Size Growth



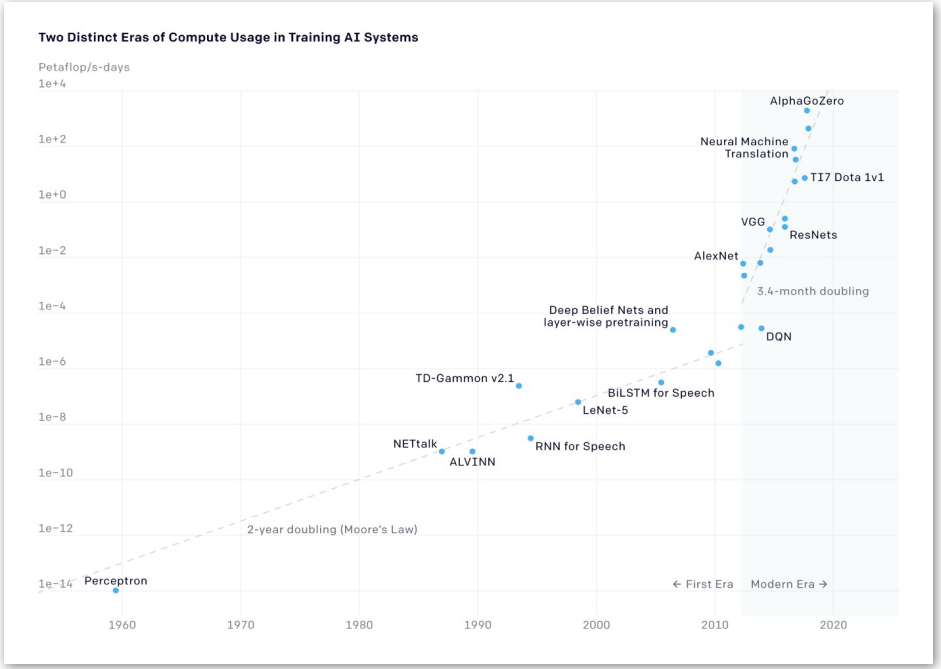
ML Compute Needs (2012 to Present Day)



Source: https://openai.com

In recent years, **computing needs grew by 300,000x** to train the machine learning models that are widely deployed in the industry

ML Compute Needs (from the 1960s)



Source: https://openai.com

In recent years, the amount of computing needed has grown remarkably fast.

Compute requirements are **doubling nearly every 3 to 4 months**

Cloud TPU



Source: Google



Source: Google



Source: Google



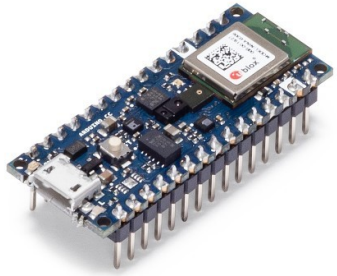
Source: Google

Cloud TPU

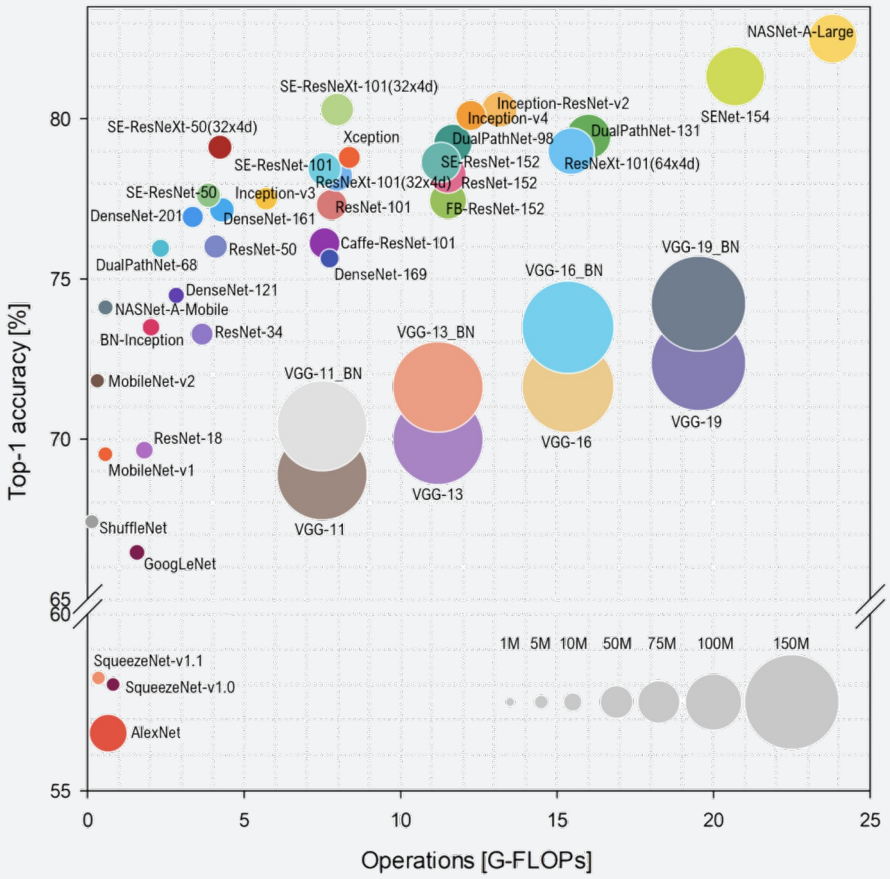




TinyML



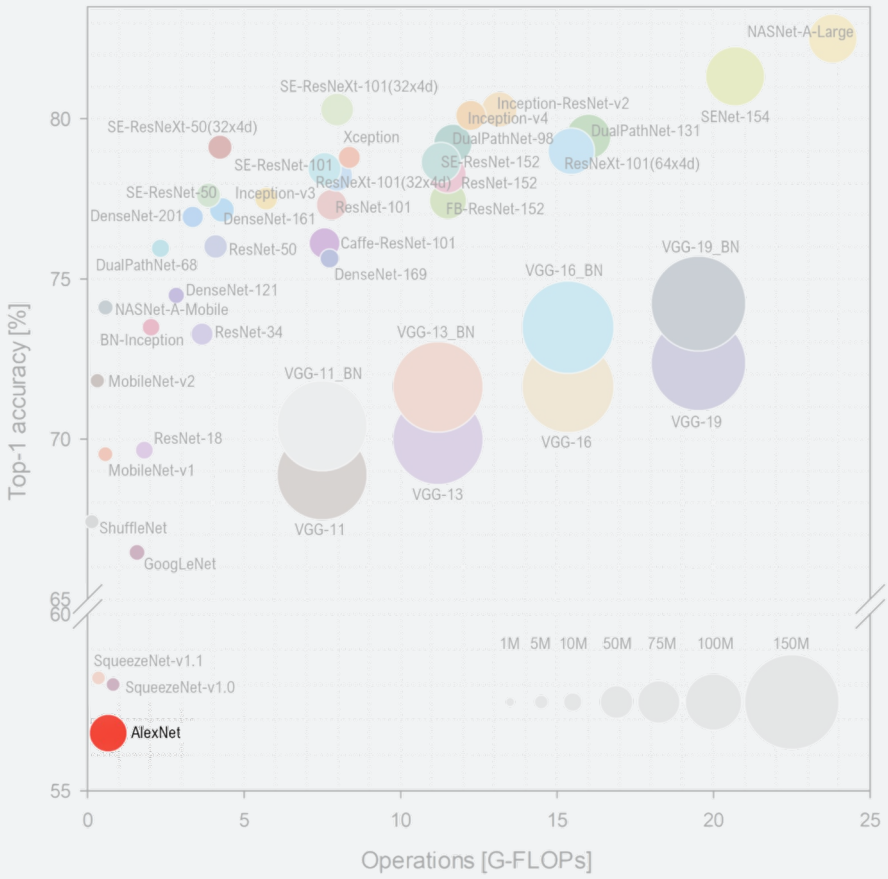
ML Model Evolution



Source: S. Bianco, R. Cadene, L. Celona, and P. Napoletano, “Benchmark analysis of representative deep neural network architectures,” *IEEE Access*, vol. 6, pp. 64 270–64 277, 2018

ML Model Evolution

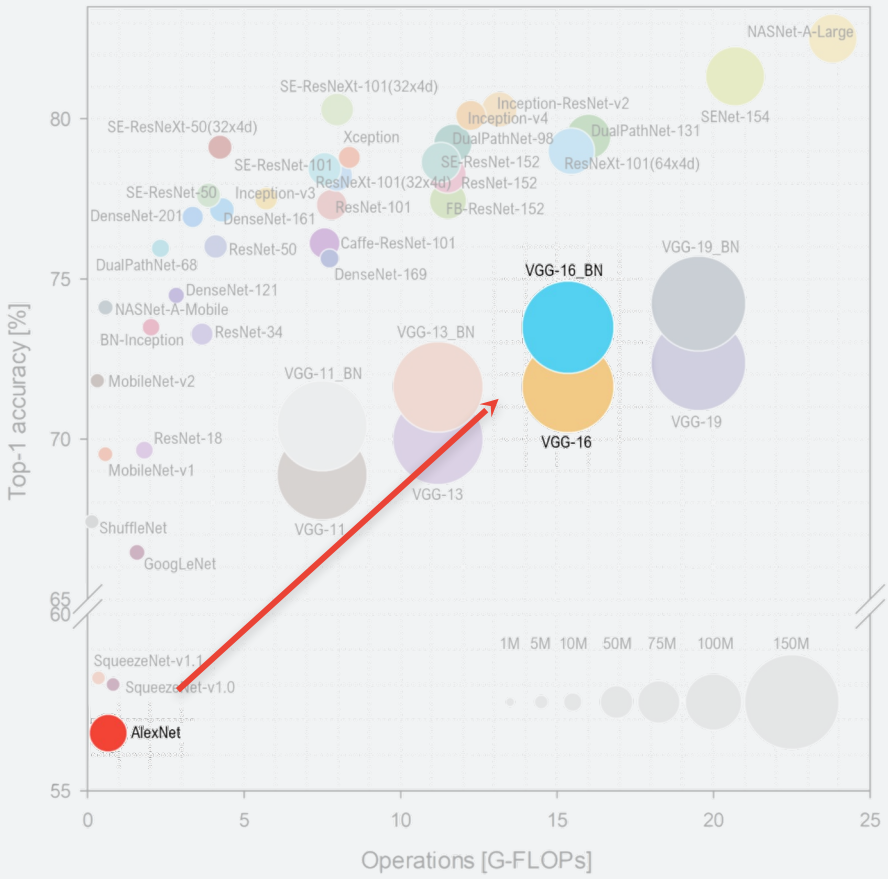
- **AlexNet** (2012)
  - 57.1% accuracy
  - 61MB in size



Source: S. Bianco, R. Cadene, L. Celona, and P. Napoletano, “Benchmark analysis of representative deep neural network architectures,” *IEEE Access*, vol. 6, pp. 64 270–64 277, 2018

ML Model Evolution

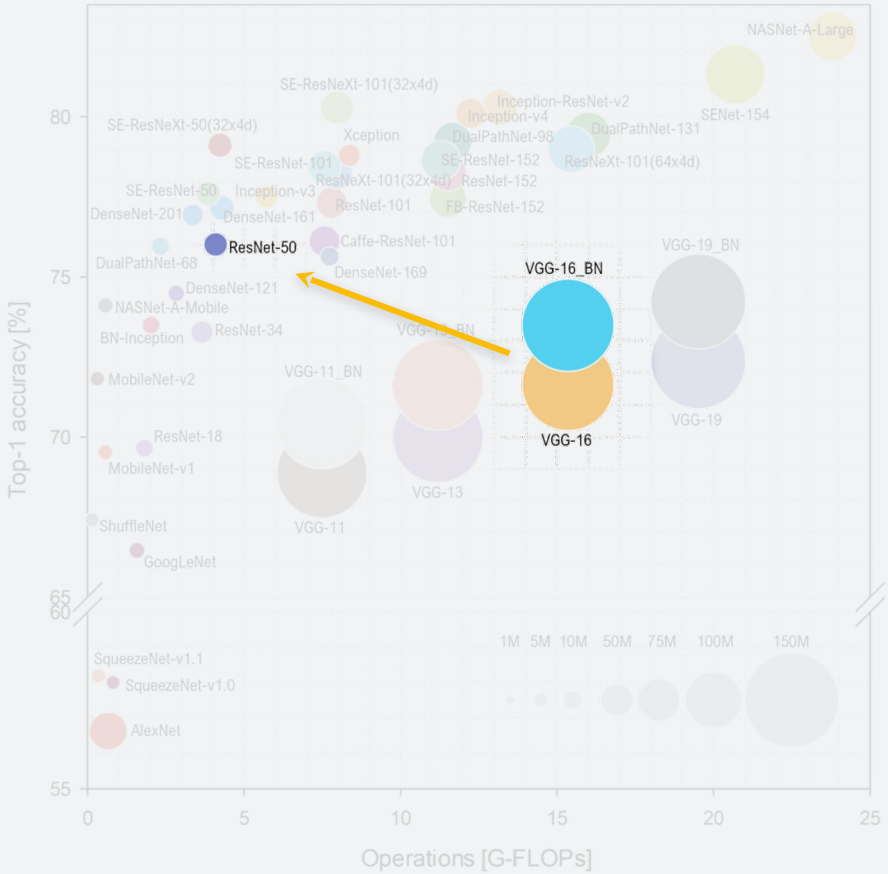
- **VGGNet** (2014) [VGG-16]
  - **71.5%** accuracy
  - **528MB** in size



Source: S. Bianco, R. Cadene, L. Celona, and P. Napoletano, “Benchmark analysis of representative deep neural network architectures,” *IEEE Access*, vol. 6, pp. 64 270–64 277, 2018

ML Model Evolution

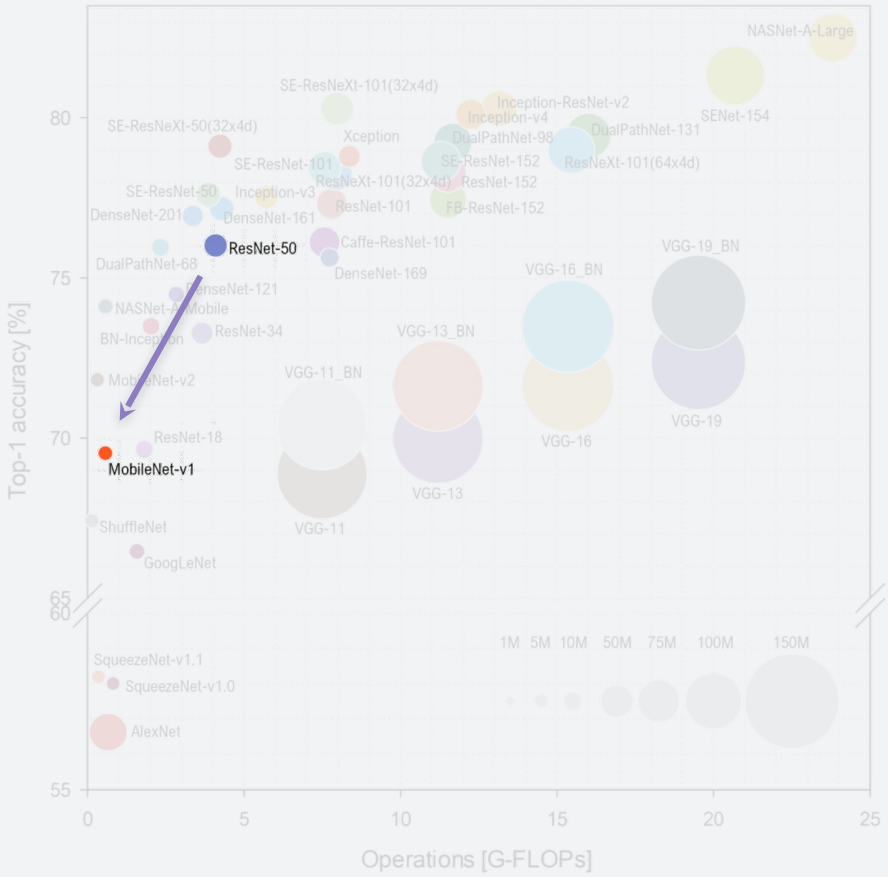
- **ResNet** (2015)
  - **75.8%** accuracy
  - **22.7MB** in size



Source: S. Bianco, R. Cadene, L. Celona, and P. Napoletano, “Benchmark analysis of representative deep neural network architectures,” *IEEE Access*, vol. 6, pp. 64 270–64 277, 2018

ML Model Evolution

- **MobileNet** (2015)
  - ***MobileNetv1***
    - **70.6%** accuracy
    - **16.9MB** in size



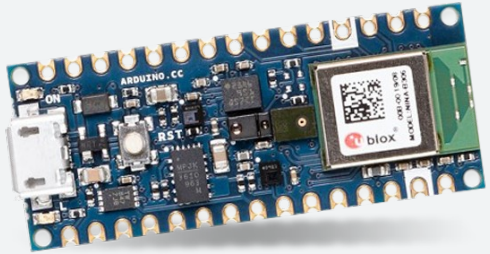
Source: S. Bianco, R. Cadene, L. Celona, and P. Napoletano, “Benchmark analysis of representative deep neural network architectures,” *IEEE Access*, vol. 6, pp. 64 270–64 277, 2018

ML Model Evolution

- **MobileNet** (2015)
  - ***MobileNetv1***
    - 70.6% accuracy
    - 16.9MB in size

Problem:

Our board (in your kit for Course 3) only has **256KB** of RAM (memory) yet ***MobileNetv1*** needs **16.9MB**!



Source: S. Bianco, R. Cadene, L. Celona, and P. Napoletano, “Benchmark analysis of representative deep neural network architectures,” *IEEE Access*, vol. 6, pp. 64 270–64 277, 2018

Fullscreen. Show presenter.